Kalman-Filtered Statistical Arbitrage

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In this report we study statistical arbitrage, seeking to exploit relationships between pairs of stocks. A set of 8 technology equities is screened for cointegration using the Engle-Granger method, and the half life of divergence from mean is computed using the Ornstein-Uhlenbeck process for cointegrated pairs. For the pair most likely in equilibrium (which at time of testing was `GOOG-AMZN`), we fit a regression model with an intercept and hedge ratio updated in time using a Kalman filter. The filter is further used to output a control variable, from which trade positions are opened and closed at fixed thresholds. Over a four year in sample period the strategy achieves a portfolio growth of £11,462.80 from a starting capital of £10,000, making a total of 73 trades. A grid search over free parameters is then performed, from which parameters are used to produce out of sample results.

I. INTRODUCTION

Pairs trading is the oldest form of statistical arbitrage. Two assets driven by a common factors are expected to maintain a long term equilibrium, and deviations from that relationship are expected to decay. The property of interest is therefore not the price of either asset, but the deviation of their relative pricing from historical norms. The relaxation time of such deviations is a key metric in determining whether or not a pair can be traded with good outcomes. The first step of this strategy is finding cointegrated pairs. In this case this means that for two non-stationary prices series $Y_t \sim I(1)$ and $X_t \sim I(1)$, we seek a hedge parameter (β) such that the residual spread $e_t = Y_t - \beta X_t - \alpha$ is stationary ($e_t \sim I(0)$).

We use the two step Engle-Granger test to test the null hypothesis that two stocks are not cointegrated, outputting a p-value being the probability that such test data can accept such a hypothesis. Alongside, it gives an estimate to the fixed hedge ratio β. The standard treatment fixes the hedge ratio β once, by ordinary least squares over the whole sample. This is unsatisfactory, it oversimplifies an ever changing market, giving too little information to make well informed trades. To combat this we fit a time series for both the hedge ratio and intercept through the use of a Kalman filter. The values of β_{t-1} and α_{t-1}, are then used to forecast Y_t and the deviation of such forecast from the realised price, $e_t = Y_t - \beta_{t-1}X_t - \alpha_{t-1}$, is the spread we trade.

Because e_t is measured in price units and its scale varies over time we standardise such quantity before using it to make trading signals. We compute a rolling mean and standard deviation of spread over a set window of time, and then compute $z_t = \frac{e_t - \bar{e}_t}{s_t}$. Trading signals can then be created using set threshold values of z_t for both entry and exit into positions with no look ahead bias.

II. METHODS

The analysis is implemented in Python using `yfinance` for data acquisition, `pandas/numpy` for numerical linear algebra, and `statsmodels` for statistical modelling.

A. Data Construction

For the tech companies 'NVDA', 'AAPL', 'MSFT', 'AVGO', 'AMD', 'GOOG', 'AMZN' and 'META' 5 years of daily closing prices are downloaded using `yfinance`, forming a matrix $P_{t,i}$, t indexing the time given by dates and i the companies. Rows containing any missing entries are dropped. Prices are converted to logarithmic prices ($p_{i,t} = \ln(P_{i,t})$, which are used in preference to simple prices because they are time additive and scale invariant. The prices are then split 80/20 for training and testing respectively. From here we will drop the matrix notation and just consider $p_{i,t} = p_i(t)$.

B. Cointegration and pair selection

Two log-price series $p_x, p_y \sim I(1)$ are cointegrated if there exists $\beta \neq 0$ such that the residual

$$\varepsilon_t = p_y(t) - \alpha - \beta\, p_x(t) \tag{1}$$

is stationary, $\varepsilon_t \sim I(0)$. The Engle-Granger two-step procedure estimates (α, β) by OLS,

$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_t \big[p_y(t) - \alpha - \beta p_x(t)\big]^2, \tag{2}$$

and then applies an augmented Dickey-Fuller test to the fitted residuals $\hat{\varepsilon}_t$,

$$\Delta\hat{\varepsilon}_t = \phi\,\hat{\varepsilon}_{t-1} + \sum_{k=1}^{K} c_k\,\Delta\hat{\varepsilon}_{t-k} + \eta_t, \tag{3}$$

testing $H_0 : \phi = 0$ (unit root, no cointegration) against $H_1 : \phi < 0$.

The test in Eq. (3) is not symmetric under $x \leftrightarrow y$, because the OLS fit (2) minimises error in y only. We therefore compute both orientations and screen on the more conservative of the two,

$$p_{xy}^{\text{screen}} = \max\big(p_{y|x},\, p_{x|y}\big), \tag{4}$$

while fitting the hedge ratio in the better-conditioned orientation $\arg\min(p_{y|x}, p_{x|y})$. A pair enters the candidate set if $p_{xy}^{\text{screen}} < p_{\text{thr}} = 0.05$.

C. Mean-reversion timescale

Evidence of cointegration is necessary but not sufficient. The spread must revert on a timescale short enough to trade. Model the spread as an Ornstein-Uhlenbeck process,

$$\mathrm{d}S_t = \theta\,(\mu - S_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \theta > 0, \tag{5}$$

with W_t a Wiener process. Equation (5) is the Langevin equation for an overdamped particle in the harmonic potential $U(S) = \frac{1}{2}\theta(S-\mu)^2$, and its conditional mean decays exponentially,

$$\mathbb{E}[S_t \mid S_0] = \mu + (S_0 - \mu)\,e^{-\theta t}, \tag{6}$$

with stationary variance $\sigma^2/2\theta$. Discretising Eq. (5) at unit time step gives the time series,

$$\Delta S_t = a + \gamma\,S_{t-1} + e_t, \qquad \gamma \simeq -\theta, \tag{7}$$

so that the reversion half-life follows from Eq. (6) as

$$\tau_{1/2} = \frac{\ln 2}{\theta} = -\frac{\ln 2}{\gamma}, \qquad \gamma < 0. \tag{8}$$

If the fitted $\gamma \geq 0$ the process is non-stationary or explosive and we set $\tau_{1/2} = \infty$, excluding the pair. The half-life sets the natural unit of time for the strategy, holding periods much shorter than $\tau_{1/2}$ exit before the potential has done its work, while lookback windows much longer than $\tau_{1/2}$ average over several independent excursions and blur the signal.

D. Dynamic hedge ratio: linear state-space model

The static fit (2) assumes (α, β) are constants of the motion. We instead promote them to a hidden state $\boldsymbol{\theta}_t = (\alpha_t, \beta_t)^{\mathsf{T}} \in \mathbb{R}^2$, observed through the price relation:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} + \boldsymbol{w}_t, \qquad \boldsymbol{w}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}), \tag{9}$$
$$p_y(t) = \boldsymbol{H}_t\boldsymbol{\theta}_t + v_t, \qquad v_t \sim \mathcal{N}(0, R), \tag{10}$$

with the observation vector

$$\boldsymbol{H}_t = \big(1,\ p_x(t)\big). \tag{11}$$

Equation (9) encodes the assumption that the economic relationship between the two equities diffuses slowly rather than erratically, Eq. (10) says that the observed log price of y equals the equilibrium level implied by x plus a disequilibrium term v_t. The covariances are parameterised as in Ref. [1],

$$\boldsymbol{Q} = \mathrm{diag}\left(\frac{\delta_\alpha}{1-\delta_\alpha},\ \frac{\delta_\beta}{1-\delta_\beta}\right), \qquad R = 1, \tag{12}$$

so that the dimensionless quantities $\delta_\alpha, \delta_\beta \in (0,1)$ control the ratio of process to measurement noise (how fast the state is allowed to move). Only the ratio $\boldsymbol{Q}/R$ is identified by the filter, which is why R may be fixed at unity without loss of generality.

Given $\boldsymbol{\theta}_{t-1}$ and its error covariance $\boldsymbol{P}_{t-1}$, the Kalman recursion [2] is the prediction step

$$\hat{\boldsymbol{\theta}}_{t|t-1} = \boldsymbol{\theta}_{t-1}, \tag{13}$$
$$\boldsymbol{P}_{t|t-1} = \boldsymbol{P}_{t-1} + \boldsymbol{Q}, \tag{14}$$

followed by the update

$$e_t = p_y(t) - \boldsymbol{H}_t\hat{\boldsymbol{\theta}}_{t|t-1}, \tag{15}$$
$$S_t = \boldsymbol{H}_t\boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^{\mathsf{T}} + R, \tag{16}$$
$$\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^{\mathsf{T}}S_t^{-1}, \tag{17}$$
$$\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_{t|t-1} + \boldsymbol{K}_t e_t, \tag{18}$$
$$\boldsymbol{P}_t = \big(\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H}_t\big)\boldsymbol{P}_{t|t-1}. \tag{19}$$

Because the state transition matrix in Eq. (9) is the identity, Eq. (13) is trivial and the prediction covariance update (14) may equivalently be applied at the end of each iteration.

The quantity of interest is the innovation e_t of Eq. (15). It is the one-step-ahead forecast error of p_y given all information up to $t-1$, and it plays the role of the spread (1) with the advantage of using no future information. The filter is seeded over a warm-up window of $n_w = 30$ days by OLS, $\boldsymbol{\theta}_0 = (\hat{\alpha}, \hat{\beta})^{\mathsf{T}}$ with $\boldsymbol{P}_0 = \widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}})$ the estimated coefficient covariance matrix, which correctly encodes the initial uncertainty and the α–β correlation.

E. Signal generation

e_t carries units of log price and its scale drifts with market volatility. It is made dimensionless by standardising over a rolling window of length w,

$$z_t = \frac{e_t - \bar{e}_t^{(w)}}{s_t^{(w)}}, \qquad \bar{e}_t^{(w)} = \frac{1}{w}\sum_{k=0}^{w-1} e_{t-k}, \tag{20}$$

with $s_t^{(w)}$ the corresponding rolling sample standard deviation.

The position $\pi_t \in \{-1, 0, +1\}$ is a deterministic function of z_t with entry threshold z_e, exit threshold $z_x < z_e$ and stop-loss $z_s > z_e$. Writing π_{t^-} for the position carried into day t,

$$\pi_t = \begin{cases} -1, & \pi_{t^-} = 0 \text{ and } z_e \leq z_t < z_s, \\ +1, & \pi_{t^-} = 0 \text{ and } -z_s < z_t \leq -z_e, \\ 0, & \pi_{t^-} = +1 \text{ and } (z_t \geq -z_x \text{ or } z_t \leq -z_s), \\ 0, & \pi_{t^-} = -1 \text{ and } (z_t \leq z_x \text{ or } z_t \geq z_s), \\ \pi_{t^-}, & \text{otherwise}, \end{cases} \tag{21}$$

where $\pi_t = +1$ denotes long y/short x and $\pi_t = -1$ denotes short y/long x. The stop-loss branch $|z_t| \geq z_s$ is the model's fail safe. A spread that keeps diverging is evidence that a pair of equities are no longer cointegrated.

F. Returns

Positions are formed on the close of day $t-1$ and held through day t, so all signal and hedge-ratio series are lagged by one step. The return of one unit of the spread portfolio is

$$r_t^{\mathrm{sp}} = \frac{\Delta p_y(t) - \beta_{t-1}\,\Delta p_x(t)}{1 + |\beta_{t-1}|}. \tag{22}$$

The denominator normalises by the gross exposure of the two legs, so that r_t^{sp} is a return on capital rather than an unscaled log price difference. With leverage L, per-unit transaction cost τ and risk-free rate r_f (annualised, 252 trading days), the net daily return is

$$r_t = \underbrace{L\,\pi_{t-1} r_t^{\mathrm{sp}}}_{\text{trading}} + \underbrace{\frac{r_f}{252}[\pi_{t-1} = 0]}_{\text{idle cash}} - \underbrace{L\,\tau\,|\pi_{t-1} - \pi_{t-2}|}_{\text{costs}}, \tag{23}$$

and the portfolio grows as

$$V_t = V_0 \prod_{u \leq t} (1 + r_u), \qquad V_0 = \$10{,}000. \tag{24}$$

G. Performance metrics

Let $\bar{r}$ and s_r denote the sample mean and standard deviation of $\{r_t\}$. The annualised Sharpe ratio is

$$\mathrm{SR} = \frac{\bar{r} - r_f/252}{s_r}\sqrt{252}. \tag{25}$$

The drawdowns and its extremum are

$$D_t = \frac{V_t - \max_{u \leq t} V_u}{\max_{u \leq t} V_u}, \qquad \mathrm{MDD} = \min_t D_t, \tag{26}$$

and with the compound annual growth rate

$$g = \left(\frac{V_T}{V_0}\right)^{252/T} - 1, \tag{27}$$

where T is the number of years, the Calmar ratio is $\mathrm{Calmar} = g/|\mathrm{MDD}|$. We also report the daily win rate and number of trades.

H. Parameter search

Combinations of the free parameters $(z_e, z_x, w, \delta_\beta, z_s)$ using an exhaustive grid search, are tested for both the maximum Sharpe ratio and the maximum PnL (whilst limiting $MDD > -10\%$). Parameter values are as given in table I. Both are fitted on training data and evaluated on test data.

TABLE I. Parameter grid searched in-sample. All other parameters are held at $\delta_\alpha = 10^{-4}$, $R = 1$, $n_w = 30$, $L = 2$, $\tau = 10^{-3}$, $r_f = 4.5\%$.

Symbol	Grid values	Count
z_e (entry)	1.25, 1.50, 1.75, 2.00, 2.25	5
z_x (exit)	0.00, 0.25, 0.50	3
w (z-window, days)	20, 30, 45, 60	4
δ_β	10^{-5}, 10^{-4}, 10^{-3}	3
z_s (stop)	3.0, 3.5, 4.0	3
Total combinations		540

III. IMPLEMENTATION

A. Pair screening

`pair()` iterates over `itertools.combinations` of the log-price columns, calls `statsmodels.tsa.stattools.coint` in both orientations, and retains the larger p-value as the screening statistic per Eq. (4) while fitting (α, β) in the orientation with the smaller p-value. The static spread (1) is then passed to `calculate_half_life`.

B. Half-life estimation

`calculate_half_life()` regresses ΔS_t on S_{t-1} with an intercept [Eq. (7)], aligning the differenced and lagged series in a joint `DataFrame` before dropping `NaN`s so that the two regressors share an index. The condition `gamma < 0` enforces the stationarity condition below Eq. (8).

C. Kalman filter

`kalman_filter()` implements Eqs. (15)–(19) in a single pass, with two design choices worth noting. First, $\boldsymbol{Q}$ is added to $\boldsymbol{P}$ *after* the update rather than before it. Since the transition matrix is the identity this is algebraically identical to Eq. (14) applied at the next iteration, and it saves one matrix copy per step. Second, $\boldsymbol{P}$ is explicitly symmetrised, $\boldsymbol{P} \leftarrow (\boldsymbol{P} + \boldsymbol{P}^\top)/2$, to suppress accumulation of floating point asymmetry over $\mathcal{O}(10^3)$ iterations. The routine returns α_t, β_t, e_t, z_t and a terminal state dictionary $\{\boldsymbol{\theta}_T, \boldsymbol{P}_T\}$ used to seed the out-of-sample run without re-fitting at the train/test boundary.

D. Signals, returns and search

`signals()` is a direct form of Eq. (21) as a sequential loop, which is necessary because π_t depends on π_{t-1} and therefore cannot be vectorised. `returns()` applies `.shift(1)` to both π and β before computing Eq. (22),

TABLE II. Cointegrated pairs on the training window, $\binom{8}{2} = 28$ tests, screened at $p_{\rm thr} = 0.05$ using the conservative statistic of Eq. (4). β is the OLS hedge ratio of Eq. (2) and $\tau_{1/2}$ the half-life of Eq. (8).

Pair (y–x)	p-value	β	$\tau_{1/2}$ (days)
`GOOG`–`AMZN`	0.0013	0.7722	18.3
`MSFT`–`META`	0.0292	0.3935	33.2
`MSFT`–`NVDA`	0.0357	0.2597	27.6

TABLE III. In-sample performance, `GOOG`–`AMZN`, default parameters, $V_0 = \$10{,}000$, $L = 2$, $\tau = 10^{-3}$, $r_f = 4.5\%$.

Quantity	Value
Net P&L	\$11,462.83
Terminal value V_T	\$21,462.83
Sharpe ratio, Eq. (25)	0.70
Calmar ratio	0.62
Maximum drawdown, Eq. (26)	−35.42%
Daily win rate	53.32%
Total trades	73

which is the single most important line in the backtest, without it the strategy would trade on information generated by the same day's close and the results would be meaningless. `parasearch()` builds the grid with `itertools.product`, re-runs the full filter-signal-return chain for every combination of parameters, and returns one row of metrics per combination.

IV. RESULTS

A. Pair selection

Table II lists the pairs that survive the $p < 0.05$ screening in the training data. `GOOG-AMZN` is the strongest candidate in terms of p-value and also has the shortest reversion time, $\tau_{1/2} = 18.3$ days. This half life is short enough as to give us some meaningful results over the 1 year test period. Therefore, it was chosen to be the pair carried forward.

B. In-sample performance

With default parameters ($z_e = 1$, $z_x = 0.25$, $z_s = 3$, $w = 30$, $\delta_\alpha = \delta_\beta = 10^{-4}$) the strategy is in an active trade position on 617 of 974 days (303 long, 314 short). Summary statistics are given in Table III and the equity curve in Fig. 1.

The win rate barely exceeds 50%, which is expected as a mean-reversion strategy earns from a small positive skew in a large number of nearly symmetric daily outcomes, not from directional accuracy. The drawdown of −35% is the headline weakness, with $L = 2$ and a single pair there is no diversification, so one prolonged divergence dominates the risk profile.

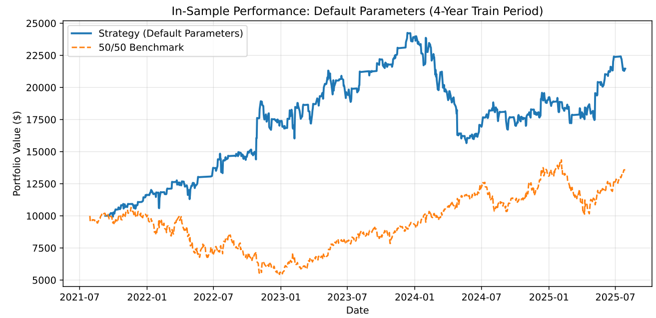

FIG. 1. In-sample equity curve V_t [Eq. (24)] against the equally weighted buy-and-hold benchmark over the four-year training window.

TABLE IV. Optimal parameter sets selected in-sample by maximum Sharpe ratio and PnL, and their out-of-sample performance.

	Max-SR set	Max-P&L set	Benchmark
z_e	2.25	2.25	—
z_x	0.25	0.5	—
w	45	45	—
δ_β	0.0001	0.001	—
z_s	3.5	3.5	—
Train SR	1.29	1.25	0.26
Test SR	0.13	1.22	1.08
Test PnL	\$453.83	\$1,338.21	\$3,487.05
Test MDD	-14.11%	-5.36%	-19.59%

C. Out-of-sample performance

The key observation is the gap between in-sample and out-of-sample performance, our portfolio no longer out grows the 50/50 benchmark. Optimisation over all 540 possible combinations of parameters improves the in-sample objective while degrading out-of-sample results. Summary statistics are given in table IV and the equity curve in fig 2

V. CONCLUSIONS

This framework tests for cointegration, returns an Ornstein-Uhlenbeck and a an adaptive hedge ratio from a Kalman filter. Generating trading signals and performing an out-of-sample backtest.

The filter tracks the hedge ratio using only past information, producing meaningful trading signals and the

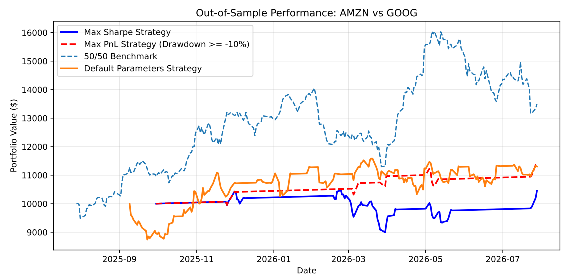

FIG. 2. Out-of-sample equity curves for both sets of parameter in table IV and defualt parameters, against the buy-and-hold benchmark, over the test data.

screening can successfully predict whether or not two pairs are cointegrated over training data. With default parameters, the in-sample portfolio significantly out performs that of the 50/50 benchmark and approximately doubles the starting capital. Which could be considered a success except for the fact used the same data to determine what pairs would the best for such trading system.

The out-of-sample results are far less impressive but they still hold some significant results. Whilst portfolio growth has reduced significantly both strategies significantly cut down the maximum drawdown and the number of trades. The PnL strategy also maintains an acceptable Sharpe ratio and consistently makes profitable trades whilst the Sharpe strategy is dominated by a losing trade. The default parameters grow portfolio roughly the same as the maximum PnL strategy even though it makes a significant losing trade at the beginning of the year, losing over 10% of the starting capital straight away. These results show that the in-sample single window parameter backtest does not generalise well over test data, and improvements are mentioned in detail in section VI. However, at a foundational level it does show how we can build upon results to create a highly profitable and low risk trading system.

VI. LIMITATIONS & IMPROVEMENTS

a. Multiple Pairs The most powerful and significant improvement by far, would be to trade a spread of pairs, eliminating down time in the strategy. It further allows the choice to trade the pair most out of equilibrium (relative and under a certain stop loss threshold), which in turn could enhance portfolio growth. This step is almost indicative to this method, as through our grid search we have reduced risk to a point where trades only happen a couple times a year, so most of the time our capital is sitting idle, reducing our potential portfolio growth significantly.

b. Estimate $\boldsymbol{Q}$ *and* R *rather than tuning them.* $\delta_\alpha, \delta_\beta$ and R are currently grid-searched. Expectation–maximisation or direct maximisation of the Gaussian log-likelihood

$$\ln \mathcal{L}(\boldsymbol{Q}, R) = -\frac{1}{2}\sum_t \left[\ln 2\pi S_t + \frac{e_t^2}{S_t}\right] \tag{28}$$

would fit the noise covariances to the data in a principled way.

c. Optimal thresholds The thresholds (z_e, z_x, z_s) are chosen by grid search, but for an Ornstein-Uhlenbeck spread with known θ and σ the entry/exit levels that maximise expected return per unit time can be derived analytically as a free boundary problem [3].

d. Position sizing. $\pi_t \in \{-1, 0, +1\}$ discards the magnitude of $|z_t|$. Continuous sizing $\pi_t \propto -z_t$ would deploy capital in proportion to the strength of the signal.

e. Halting trades Add a time stop at a multiple of $\tau_{1/2}$ (a spread that has not reverted in $3\tau_{1/2}$ is evidence against the model), and a rolling re-test of cointegration that de-lists a pair whose p-value decays. Further, if the z-score exceeds z_s, implement another threshold to resume trading signals instead of reentering as soon as the z-score drops below z_s.

f. Bias The 8 equities were chosen with the benefit of hindsight, as names that are large and liquid today. A point-in-time index membership file would remove this bias.

[1] E. P. Chan, *Algorithmic Trading: Winning Strategies and Their Rationale* (Wiley, Hoboken, NJ, 2013).
[2] R. E. Kalman, Journal of Basic Engineering **82**, 35 (1960).
[3] W. K. Bertram, Physica A **389**, 2234 (2010).